

TrioVibe: Integrated Multimodal Intelligence For Depression Detection

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Department of Computer Science

In

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Bachelor's / Master Degree in Computer Science

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This is to certify that the project titled “**TrioVibe**” is the genuine work carried out by **M Sohail Shahzad & M Mateen Mujahid**, student of BSCS of Computer Science Department, Lahore Garrison University, Lahore during the academic year 2022-26, in partial fulfilment of the requirements for the award of the degree of Bachelor of Computer Science and that the project has not formed the basis for the award previously of any other degree, diploma, fellowship or any other similar title.

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DECLARATION

This is to declare that the project entitled “**TrioVibe**” is an original work done by undersigned, in partial fulfilment of the requirements for the degree “Bachelor of Science in Computer Science” at Computer Science Department, Lahore Garrison University, Lahore.

All the analysis, design and system development have been accomplished by the undersigned. Moreover, this project has not been submitted to any other college or university.

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DEDICATION

It would be a matter of huge pride and honor to dedicate our work, research and project to our inspiration, role model, our Beloved, Honorable Holy Prophet Hazrat Muhammad. Moreover, we want to dedicate our project to our respected, beloved parents, it was their prayers which made it possible for us to come along at this stage. Besides, I devote this undertaking report to every one of my educators and companions who have helped me all through my vocation. I'm thankful to my folks for being generally so grateful. Finally, to the Lahore Garrison University and especially the department of Computer Science

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Before anything we cannot thank enough, The Almighty, The Creator, The Alpha & Omega, The Most Gracious and the Most Merciful Allah (SWT) for his countless, unlimited blessings on us, secondly all salutations are presented to our beloved Prophet Hazrat Muhammad who helps us understand our creator and whose example is the finest for us. During this task, we got backing, help or functional assistance from numerous individuals, and it is pleasing to recognize their goals. Multiple people have supported us for the completion of this project, so we would like to acknowledge them for their contribution. First of all, our respected parents without their prayers, who truly adores us and consistently appeals to God for us in each snapshot of our life. Next our supervisor Naila Noreen, who pushed, helped, guided, and supported us in our project became our north star, whom we looked upon as they have equally worked hard with us and on us. Further, Naila Noreen was there for us when we needed her and has invested her efforts, precious time and actively contributed and supported the research. A bundle of thanks to all of these honorable mentions who actively participated in our project we cannot thank them enough this would never have been possible without these people. Thank You.

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List of Abbreviation

Abbreviation	Full Form	Abbreviation	Full Form
SRS	Software Requirement Specification	CRUD	Create, Read, Update, Delete
MVC	Model View Controller	LINQ	Language Integrated Query
ERD	Entity Relationship Diagram	PDF	Portable Document Format
DFD	Data Flow Diagram	CSS	Cascading Style Sheets
UML	Unified Modeling Language	HTML	HyperText Markup Language
UI	User Interface	HTTP	Hypertext Transfer Protocol
UX	User Experience	PBKDF2	Password-Based Key Derivation Function 2

ABSTRACT

Depression and anxiety are very common these days, but a lot of people still struggle to talk about their feelings and ask for help. Things like social stigma, not knowing enough about mental health, and not being able to easily see a therapist often stop people from getting the help they need. Old ways of getting help, like in-person therapy, can be tough because of privacy worries, character assassination, and the fear of being judged. TrioVibe wants to fix this by making a smart and cool web app that handles this problem better than anyone else, letting you understand your emotions while keeping your privacy safe. Unlike other apps, TrioVibe figures out emotions by looking at your text, voice, and even facial expressions, so you can really get a clear picture of how you're feeling. The main reason we made this product is to make mental health support easy for everyone who needs it, all while keeping their privacy secure so they can get better without holding back. This product mixes health with AI to help people with mental health issues they might not want to talk about. The app has a voice assistant called AI-Pal that chats with users like a friend, offering guidance and emotional support. If someone needs clinical help, the platform can connect them with psychologists through chat and video calls. TrioVibe wants to make good mental health easy, accessible, affordable, and effective by combining AI-powered peer support with professional medical access. The purpose of this project is to create a platform which is safer and easily accessible where people can easily detect the state of their mental health and take appropriate steps to be in good shape. There are still many people living in this world who do not express their feelings and emotions with society. This product will help these people to keep their mental and emotional state in check. The main aim of the product is to make things easy and cheap so anyone can have access to improve their well being.

INTRODUCTION

In today's world, depression has significantly increased but still many people find it very difficult to share their feelings with others. They also hesitate to reach out for help. Social inequality and very less access to mental health guidelines stop individuals from getting the help they require. Previously used methods of seeking help, such as in-person therapy, can be challenging due to privacy concerns, character assassination and the fear of judgment. TrioVibe aims to fill this gap by developing an intelligent and attractive web application that takes care of that problem better than any other allowing both understanding emotion while maintaining the privacy of users. Unlike typical applications, TrioVibe detects emotions through a multi-modal approach analyzing text, voice, and facial expressions enabling users to gain objective insights into their mental state. The main motive behind developing this product is to make mental health wellness easily accessible to everyone in need while ensuring their privacy to help them overcome their issues without hesitating. This product is a project to blend health with artificial intelligence, helping users overcome mental health issues that they may struggle to express to others. The app contains a voice assistant named AI-Pal, it talks to users like a friend to give guidance and emotional support for those users who are in need for clinical inference, the platform does have a functionality to connect physical psychologists with those in need of emotional support through chat and video call sessions. By joining AI-powered peer support tools with professional medical access, TrioVibe aims to make good mental health easy, accessible, affordable, and effective. This product is developed as a mix of mental health with artificial intelligence to help users in overcoming their mental health issues that they face and can't express in front of other people. By detecting emotions through text, voice, and facial expressions, it helps users to better understand their emotional state. After understanding their emotions and knowing their mental health state, the user can use the voice assistant which helps the user improve their emotional state by talking with it. It talks to the user as a friend which will help him overcome his issues and express himself openly.

LITERATURE REVIEW

Table 1

Method / System	Technique / Approach	Advantages	Disadvantages / Gaps	TrioVibe Solution
Traditional Clinical Therapy [6]	Face-to-face sessions and manual PHQ-9 questionnaires.	High clinical accuracy and human touch.	High cost, geographical barriers, and social stigma.	Provides a digital gateway that reduces stigma.
Mood Tracking Apps (e.g., Daylio) [7]	Manual user logging and self-reporting.	Easy to use; helps identify long-term patterns.	Users may forget to log or provide inaccurate data.	Auto-detects emotions via Face, Voice, and Text without relying on manual logs.

Rule-Based Chatbots (e.g., Woebot) [8]	Decision trees and scripted responses.	Available 24/7; provides basic coping tools.	Scripted nature feels clinical; no multi-modal awareness.	Uses generative LLMs for natural, empathetic conversation with multi-modal awareness.
Single-Modal AI Systems [9]	Specialized NLP or Computer Vision only.	High accuracy in one specific area.	Fails to cross-verify emotions across different channels.	Uses Text for emotional score.
Tele-Health Platforms [10]	Standard Video Calling (Zoom/Skype).	Connects patients to doctors remotely.	no built-in AI tools for emotional monitoring during the session.	Combines AI monitoring, therapy marketplace, and WebRTC in one platform.

PROBLEM DEFINITION

In the modern era, mental health issues face a demanding hurdle distinguished by an increasing gap between the rising normality of mental disorder and the reach of readily accessible clinical support. The usual ways we diagnose things rely on interviews and self-perception tools like the PHQ-9, which can be skewed by people trying to look good. This leads to bad diagnoses, and the system itself is expensive and hard to access, making it a luxury. Even when resources are there, stigma makes them invisible, making getting help harder than the illness itself. Current digital tools seem promising but are often just conditional statements that can't really understand human distress. TrioVibe bridges this gap with a smart engine that combines text sentiment, voice tone, and facial biometrics to create a high-quality profile, getting around the problems of traditional self-reporting. This tech approach gets rid of cost and stigma by offering a private, immediate, AI-powered first line of defense. By combining this automated support with a professional marketplace, the project solves the major issue between initial emotional awareness and clinical recovery, making sure that users have an easy path for detecting problems to receive the solutions.

SOFTWARE REQUIREMENT SPECIFICATION

1. Introduction

1.1 Purpose

The purpose of this project is to create a platform which is safer and easily accessible where people can easily detect the state of their mental health and take appropriate steps to be in good shape. There are still many people living in this world who do not express their feelings and emotions with society. This product will help these people to keep their mental and emotional state in check. This product is developed as a mix of mental health with artificial intelligence to help users in overcoming their mental health issues that they face and can't express in front of other people. By detecting emotions through text, voice, and facial expressions, it helps users to better understand their emotional state. After understanding their emotions and knowing their mental health state, the user can use the voice assistant which helps the user improve their emotional state by talking with it. It talks to the user as a friend which will help him overcome his issues and express himself openly. Some users still need more psychological support, they chat and make video calls with the verified psychologists. The main aim of the product is to make things easy and cheap so anyone can have access to improve their well being.

1.2 Document Conventions

The document is following the IEEE Standard for Software Requirements Specification (IEEE 830-1998). The document is styled and formatted in such a manner, so that it can be easily readable and shows professionalism. The font size used for headings is 13. The font style that is used in the SRS is Times New Roman. The body of the document has a font size of 12. The line spacing is set to 1.5 to make the document readable.

1.3 Intended Audience and Reading Suggestions

The intended audience for this application includes both the ones who need help and the ones who can help them, such as professional therapists. The users need some emotional help in understanding their emotions and mental health. In case the users are facing some psychological problems like stress, anxiety and depression, this platform provides an easy and private way for them to deal with these problems. These users interact with the app's features like emotion detection, AI Pal (voice assistant), breathing exercises, and progress tracking which will help them according to their need. The application is developed in such a way that the user doesn't need technical knowledge to use it which will allow people from different age groups and diverse backgrounds to use it. The other users that serve as intended audience for this project are the Psychologists and counselors. They are the users that help the users that are there in application for professional guidance and emotional support. These users can chat with patients that need psychological support and make video calls for better support. These users also don't have to require technical knowledge to use this application. They are basically here for the emotional support and therapy of the patients. Both users are essential as the patient requires psychological support, and the psychologists are here for guidance and support.

It is suggested that the SRS be read in its entirety to gain a full understanding of the requirements for the project. It is organized as follows:

Section 1: Introduces the document and the product.

Section 2: Provides an overall description, operating environment etc. of the product and its functions.

Section 3: Specifies the external interface requirements of the system.

Section 4: Specifies the system feature, stimulus/ response and functional requirements of the product.

Section 5: Lists any nonfunctional requirements, safety and security features and business rules that the product must follow.

1.4 Product Scope

TrioVibe is a web-based application designed to make mental health care more accessible and comfortable for everyone. It focuses on helping people manage depression, stress, and emotional struggles by detecting their emotions through text, voice, and facial expressions. The system uses advanced AI models to detect the current emotion of the user. The most interesting feature of this application is the voice assistant which helps the user by conversing with them in a friendly manner providing comfort. Another aspect is the real-time feedback and guidance which the assistant provides. The support doesn't end here with the voice assistant if the user still needs further assistance, then their chat functionality where the user can chat with the professional psychologist and make video calls with it. The combination of artificial intelligence and the health experts create a platform which will promote human well-being. This product removes common problems like costly therapy, accessibility and waiting times. It makes the platform where mental health is easily available and accessible in a safe and private manner.

2. Overall Description

2.1 Product Perspective

TrioVibe is a web-based platform designed to identify and understand human emotions through artificial intelligence. It is built using the MERN stack and provides a uniform, fast, and interactive experience for users which need emotional awareness and mental health support. The system gets input from the user through text, voice and facial expression. It detects the emotion of the user and displays it on the screen which shows the user their current mental state. There is an AI assistant in the application which is known as AI Pal to provide comfortable conversation to the user. If the user still requires further assistance, then there is functionality which is used by the user to contact the physical psychologist through application. The application is primarily developed to make mental health easily available by combining the artificial intelligence approach and other functionalities.

2.2 Product Functions

The product has several functions that are defined below:

2.2.1 Emotion Detection

TrioVibe can detect emotions on textual data, voice data or facial expressions by using the ML trained models which allows apps to detect the emotion of the user like happy, sad and anger etc. The data obtained through input are used by the system to detect the emotion of the user and provide them with their emotional state. This will allow the user to know about their mental health and take steps according to results.

2.2.2 AI-Based Psychological Support

There is a voice based assistant in the application called AI Pal it gets voice inputs and acts as a therapy bot to interact with the user. It has a friendly nature through which the user can interact and get emotional insights. It receives input through voice and converts it into text and sends it to google LLM models which in turn gives responses that are voice based and friendly. This feature allows the user to express their feelings openly without any hesitation and in a private way.

2.2.3 Chat with Psychologists

TrioVibe allows the user that needs further special assistance with mental health to directly interact with the psychologist through the chat app which is also implemented in the application. This will provide an environment for the user to share their problems and get personalized feedback through physical psychologists.

2.2.4 Video Call Sessions

The user that needs further special assistance can connect with a psychologist through the application. They can book appointments with psychologists through chat app. There is a video call feature implemented in the application which is used to have a live session between the user that needs assistance and the psychologist. Its main feature is that the psychologist can see the user's facial emotion through facial emotion detection for better assistance. This will promote end-to-end encryption between the user and the psychologist.

2.2.5 User Management and Data Security

The application has implemented user management functionality providing a safer environment for the user. The authentication is implemented in the app so that only authorized users have only access to their information. The interaction history and the detected emotion of the user is stored in the database to enable the user to continue where the user left off. It will maintain the confidentiality of users.

2.3 User Classes and Characteristics

The application is designed in such a way that the users with no technical background can use the application easily. The different users according to their usage are:

i. Regular Users

- These are the users which need emotional assistance.
- They use features like chatting, emotion detection, breathing exercises, and progress tracking.
- No technical skills required.
- **Priority:** High

ii. Psychologists / Counselors

- Use the app to check user progress or emotional trends.
- Required less technical knowledge.
- **Priority:** Medium

iii. Administrators

- MIt manages user accounts and system performance.
- Data protection and security updates are handled by the admin.
- **Priority:** Medium

2.4 Operating Environment

This environment on which this app can be operational are:

- **Operating System (OS):** It can be run on windows, linux, and macOS.
- **Technology Stack:** MERN (MongoDB, Express.js, React.js, Node.js).
- **Dependencies:** This app depends on the pretrained model for facial detection, Grok for voice assistance, and web sockets for chat features.
- **Hardware Requirements:** Doesn't require any hardware as such, Just require a microphone and web camera.
- **Browser Support:** Google Chrome, Mozilla Firefox, and Microsoft Edge.

2.5 Design and Implementation Constraints

i. Design Constraints:

- Emotion detection models require processing power for real time analysis of text, voice, and video.
- The application requires stable internet connection for AI Pal voice communication and video calls.
- User data must remain secure and comply with standards that are defined for mental health applications.
- The application uses the MERN stack for web and uses the Grok API for Ai based counseling.

ii. Implementation Constraints:

- The application can be run on Node js version 18 or higher and also React version 18 or higher.
- It requires a camera and microphone for emotion detection.
- Database normalization should be performed for handling real time features.
- The application should follow secure communication using HTTPs.

2.6 User Documentation

The application will include clear documentation to help users understand and use TrioVibe easily.

The user will be guided on how to:

- Create an account and log in securely.
- Use text, voice, and facial input for emotion detection.
- Interact with the AI Pal for instant support.
- Chat with psychologists or schedule a video call.
- Manage personal profiles and view past sessions.

2.7 Assumptions and Dependencies

i. Assumptions:

- Users will have access to a stable internet connection for real-time emotion detection and video calls.
- The users will allow access to their webcam and microphone for voice and facial analysis.
- Psychologists using the system are verified professionals.

ii. Dependencies:

- The application will depend on pretrained AI models for emotion detection through text, voice, and face.
- The system will require a voice assistant which is known as AI Pal for psychological support and therapy.
- The application will use MERN stack for its website functionality.
- The application will rely on WebRTC & Ice for chat and video communication features.

We can better plan for the application's development and implementation if we comprehend these assumptions and dependencies.

3. External Interface Requirements

3.1 User Interfaces

The application's interface will be easy to use, with smooth page-to-page navigation. It will be responsive and work well on both desktop and mobile devices. The main screen consists:

- **Login Page:** A secure gateway for returning users to resume their sessions through secure credentials.
- **Sign-Up Page:** A streamline interface allowing new users to register themselves on the platform and join the TrioVibe ecosystem.
- **Home Page:** A central nervous system of application, collection of all the core features and diagnostic tools into a single dashboard.
- **Profile Update Page:** A dedicated space allowing the user to change their identity and refine their online presence.
- **Chatting Page:** A real time interface designed to create peer-to-peer connection and break the cycle of isolation.
- **Emotion Detection Page:** Shows results from text, voice, and facial emotion analysis.
- **AI Pal Page:** Offers AI-driven conversations for emotional support.
- **Breathing Exercise Page:** Guides users through calming breathing activities.
- **Progress Page:** Displays user progress and emotional trends over time.
- **Questionnaire Page:** Collects user feedback and mental health assessments.

The main goal of this design is to be easy to use and clear, providing better experience for the users.

3.2 Hardware Interfaces

In order to effectively perform its functions, this application connects with a number of hardware components. These include:

- **Devices:** The application can only be accessed on a computer or mobile device that has a modern web browser.
- **Internet Access:** An internet connection is required to run this application.
- **Webcam and Microphone:** The microphone and camera is also required to use this application
- **Memory and Processor:** As it is a web application, it can be easily run on any smart device with browsing capability.

To guarantee availability and accessibility, the application is made to work well with standard hardware.

3.3 Software Interfaces

The application requires many software components combined to work properly. These include:

- **MongoDB:** It is used as a database to store the user information, emotions and their usage history.[11]
- **Express.js and Node.js:** It serves as a main component between frontend and backend. It handles HTTPs requests.
- **React.js:** The front end is developed using it to make it more appealing and interactive.
- **Pretrained AI Models:** These are used for the detection of user emotion using text, voice and face.
- **Grok API (AI Pal):** It is a voice assistant which provides psychological support to the user.
- **WebSocket and WebRTC:** The chat app is developed using webRTC and WebSocket which allows video calling and texting functionality.

3.4 Communications Interfaces

- **Internet:** An internet connection is required for the app to perform its functions like emotion detection, voice assistant, chat and video calling.
- **Localhost HTTP Server:** The application will run locally and is accessible via `http://localhost:5173` on the user's web browser.
- **WebSocket and WebRTC:** Used for real-time messaging and video conferencing between users and psychologists.
- **Server Access:** Node.js allows JavaScript to run outside the browser and its contents can be easily displayed using a web browser.
- **Data Security:** To preserve user privacy and maintain confidentiality of psychological sessions, all communications are encrypted.
- **Optional Cloud Storage Integration:** The application can also include the cloud server and storage for easy access over the internet. Cloud services that will be used for it are AWS, Azure and Google cloud storage. HTTP's protocol will be used for the transfer of data using these services.

4. System Features

4.1 Emotion Detection

4.1.1 Description and Priority

- **Description:** Pretrained AI models enable the application to detect emotions in text, voice, and facial expressions which are used to detect the user's emotion and emotional state.
- **Priority:** High

4.1.2 Stimulus/Response Sequences

- Input can be taken using text, voice and video.
- To process the input, the application uses pre pre-trained AI model.
- Data is being processed and user emotion health is detected.
- After analyzing the information, the emotion is detected and displayed on the screen.

4.1.3 Functional Requirements

REQ-1: Application will accept voice, text, and live video input.

REQ-2: Pretrained AI models for emotion detection should be used by the application to process the input.

REQ-3: Emotion is displayed on the screen within no time.

REQ-4: If an error occurs then appropriate error messages will be displayed.

4.2 AI Pal (Psychological Assistant)

4.2.1 Description and Priority

- **Description:** With AI Pal, users can interact with an AI-powered assistant for mediocre level psychological advice and emotional support.
- **Priority:** High

4.2.2 Stimulus/Response Sequences

- The user activates the AI Pal voice assistant.
- The user then speaks to the assistant using a microphone.
- AI Pal analyzes the user's voice and responds with a calm and supportive voice reply.

4.2.3 Functional Requirements

REQ-1: The user's voice input should be heard by the AI Pal.

REQ-2: The voice should be processed and sent to the backend.

REQ-3: After that, it produces and delivers suitable voice responses.

4.3 Web Interface

4.3.1 Description and Priority

- **Description:** The program features a web interface that will make all of the key components visible. The interface guarantees responsive design for multiple screen sizes, smooth navigation, and real time updates.
- **Priority:** High

4.3.2 Stimulus/Response Sequences

- The user opens the web application in a browser.
- After that the system loads the dashboard with AI Pal and related features.
- Then the user interacts with buttons, menus, or the voice assistant.
- The system updates and displays responses instantly without refreshing the page.

4.3.3 Functional Requirements

REQ-1: Must use a responsive React-based interface to show every feature of the system.

REQ-2: It facilitates seamless switching between various sections (Home, Voice Assistant, etc.).

REQ-3: Developed in such a way that it will continue to be compatible with mobile devices and contemporary browsers.

REQ-4: If an error occurs then appropriate error messages will be displayed in the console tab.

4.4 User Authentication and History Management

4.4.1 Description and Priority

- **Description:** Users can create and access their accounts using the application. It makes sure that each user's data is securely stored and accessible at a later time.
- **Priority:** High

4.4.2 Stimulus/Response Sequences

- The user opens the app and navigates to the login or signup page.
- The user enters the data to login or registers a new account.
- The system verifies that data.
- After logging successfully, the user's previous history and saved data show in the profile.

4.4.3 Functional Requirements

REQ-1: Permit users to sign up with a password and email address.

REQ-2: Login the user by matching credentials and encryption protocols.

REQ-3: When a user logs in, they can access, view, and store their history in the database.

REQ-4: Make sure that all user data is transmitted and stored safely using accepted security procedures.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

- The program should react to user input as soon as possible, reducing any delay. After clicking, speaking, or typing, users shouldn't have to wait a long time for the system to respond making a smooth experience.
- The voice assistant used in the application should respond to the user's voice input in no time so users don't have to repeat the sentences and avoid making them feel awkward.
- The web application shouldn't take no more than five seconds to complete loading and displaying. This will attract the user and boost the traffic on the application.
- The application allows multiple users to use the application concurrently without any serious delay or slowing down.

5.2 Safety Requirements

- The information of the user should be kept safe and secure to ensure the application integrity.
- User information such as chat and login credentials is stored in such a way that no unauthorized user can access it.
- The application should be checked weekly for routine maintenance and ensure the safety of the application.
- If the user deletes their account, then all the information of that user should be permanently removed, no data would be held from the app to ensure fairness.

5.3 Security Requirements

- The authentication and authorization should be implemented in such a way that only the authorized user can have access to the system login info.
- The transfer protocol used for data transfer should be HTTPS. It prevents the user's information and data from being stolen over the network.
- The users which are only authorized should have access to the system and database. No other user will be allowed to access this information to maintain privacy.

5.4 Software Quality Attributes

- The user without having any technical knowledge can easily use the application and navigate between different features of the product.
- Responsive design should be considered for the application to run on every device.
- The system doesn't have to require any dependency to run. It should be free from errors and bugs.
- The application is designed in such a way that it is easy to navigate and users are able to access every information in the application.

5.5 Business Rules

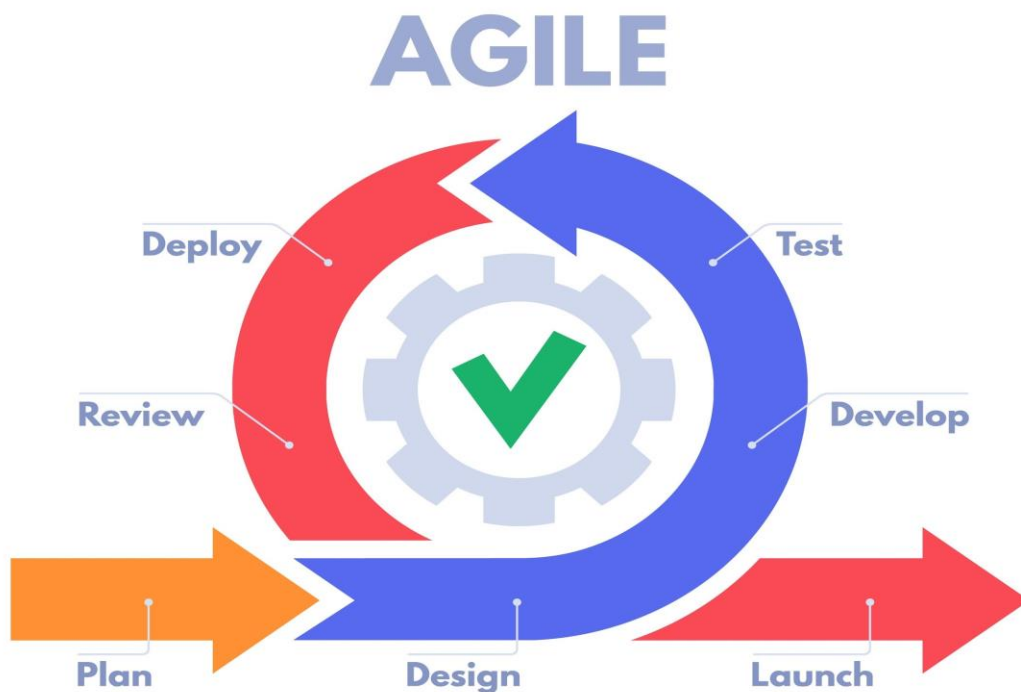
- It should only be applied to emotional support and mental health. It is strictly forbidden to use it for unethical, harmful, or illegal purposes.
- All user data should be handled responsibly, and professionalism confidentiality should be upheld by psychologists and platform administrators.
- Growth will never come at the cost of integrity, every new feature will be under the same privacy code that is of now.

METHODOLOGY

The TrioVibe development lifecycle uses an iterative framework that closes the gap between full-stack development and advance effective computing, here we break down the architectural pillars mutli-sensor data flow and integrating real time interaction protocols

5.1 Development Lifecycle

The development lifecycle used for this web app is Agile. This lifecycle allows the loop for implementation of application and feedback sprints.



5.2 Technical Architecture (MERN Stack)

The application is built using the popular web framework which is Mongo, Express, React and Node.js. This architecture boosts the performance of this app and ensures that it is accessible on every browser.

- **Frontend (React.js & Vite):** The frontend's built with React.js for a modular, component-driven feel. Vite handles the build process, giving us super-fast hot module replacement and optimized assets. For the look, we're using "Glassmorphism" with Tailwind CSS to create a modern, translucent interface that's easy on the eyes.
- **Backend (Node.js & Express.js):** The backend acts like a central nervous system, using Node.js and Express.js for critical services. It handles JWT-secured authentication and is the main gateway for lots of concurrent communication between machine learning models.
- **Database (MongoDB):** This NoSQL is used because it's flexible with schemas across different datasets, from basic user profiles to multi-modal emotion results.

5.3 Multi-Modal Emotion Detection Pipeline

The core innovation of TrioVibe lies in multi-modal analysis, it processes three different data streams to generate a unified emotional score:

- **Visual Analysis:** Using high performance pre-trained models, the platforms perform real time facial landmark extraction within the client's browser. This highly efficient advanced computing method maps small expressions to 7 primary states respectively using deep learning Convolutional Neural Network (CNN).
- **Textual Sentiment Analysis:** TrioVibe analyzes the textual data for AI-Pal interactions and personnel record, which is dispatched to Python based NLP microservices. This service uses transformer based models to parse linguistics. By evaluating the words and sense they are trying to make it differentiate them by positive or negative response and specifying the current mood.
- **Vocal Emotion Recognition:** The application captures audio performing acoustics feature extraction and speech-to-text to direct emotional indications.

5.4 AI-Pal Integration and Process Orchestration

AI-Pal a conversational interface which is powered by a highly throughput interface via Groq/GeminiApi giving contextually accurate and empathic response.

5.5 Real-Time Communication (WebRTC & Socket.IO)

For the professional therapy marketplace, the system implements:

- **Socket.IO:** We have used socket io to make sure that help is never delayed by allowing a persistent full duplex communication mode between patients and therapists.
- **WebRTC:** For video therapy TrioVibe uses WebRTC to create a private high quality digital clinic. The Node.js backend serves as a signaling server, creating the exchange of Session Description Protocol (SDP) and ICE candidates to create connection over the network, due to this architecture server load is reduced.

DETAILED DESIGN AND ARCHITECTURE

This chapter provides a brief overview of TrioVibe. The project is engineered to enhance mental health monitoring and support by multi-modal effective computing. By combining textual vocal and facial indications TrioVibe creates a highly efficient and accurate emotional baseline. Users can interact with the system via text, voice, and facial expressions to find out their real time emotional analysis, and generative engagement via an AI companion.

6.1 System Architecture

The system architecture is created as a modular ecosystem. MERN Stack is used as a bridge between web scale reliability and effective computing, this modular design allows it to grow and adapt with time. The architecture divides the application into three parts, component based frontend, centralized API organizer also known as backend and a multi-modal emotion detection engine, this separation allows the platform to concurrently handle high data streams.

6.1.1 Architecture Design Approach The design approach is rooted around Modularity, Real time Interactivity, and cross platform resilience. The goal is to achieve a clean interface that does not compromise on the performance. The architecture is divided into several key components:

- **Modular Design:** The application is designed by a series of independent living modules, like authentication, detecting emotions and a clinical marketplace. It makes sure that if you're working on or updating one part, it won't mess up the whole system.
- **Frontend Development:** The client side uses Reactjs and Vitejs for quick state management and a clean look. The user interface has a cool glassmorphism design.
- **Backend Integration:** At the heart of TrioVibe is an interprocess communication layer. The Node.js backend is like the brain, handling HTTP requests and starting Python child processes to run specific machine learning scripts for sentiment and vocal analysis.

- **Platform-Agnostic Scripting:** To ensure stability in cross platform. The backend is designed in such a way that it automatically detects the change in platform and making sure a smooth transition from development to production
- **Data Management:** MongoDB serves as a document-oriented database to store flexible schema ranging from basic user profile to journal of emotion maintaining a clean history.

6.1.2 Architecture Design: The architecture is further divided into three independent segment each segment with its own responsibilities and together they complete the TrioVibe

- **Presentation Layer**
 - **User Interface (UI):** The UI of TrioVibe allows users to input data through text, voice and video. Then it shows the detected emotions of the user as output through its attractive dashboard.
 - **Real-time Feedback:** It is built in a way to provide the user with immediate response. The user provides input data and detected emotions are shown in the result as real time.
- **Application Logic Layer**
 - **AI layer:** It handles the communication between the web App and the trained model. It gets the output from the trained model and provides it to the web app for the result. It acts as a middleman.
 - **Real time Engine:** This layer also implements the chat feature between all entities on the server and provides the real time video calling feature.
- **Data Management Layer**
 - **Secure Resolve:** Secrecy is the main motive behind this app and strong measures have been taken to keep everything confidential. The authentication is implemented through JWT to keep the data secure.
 - **History Management:** This layer includes the implementation of the history mechanism of the user in which only authorized users can access it.

6.1.3 Subsystem Architecture The architecture of this web app is further divided into several subsystems:

- **Intelligence Subsystem**

- It acts as the base system for emotion detection and gets the input from the user through text, voice and face.
- It uses logistic regression for better prediction and accuracy on textual based data and uses faceAPIs.js for facial detection.
- **AI-Pal Subsystem**
 - This subsystem includes the integration of voice assistance with the web app. It provides a better approach for assistance.
 - Provides responses on the basis of emotional state of the user using the Groq Api which only provides responses on the basis of specific parameters that are defined.
- **Communication Subsystem**
 - It basically focuses on the implementation of Therapist system in the web app by which users can book appointments from therapists.
 - After appointment booking, it implements the video call feature by which users can get therapy sessions from the therapists.

6.2 DETAILED SYSTEM DESIGN

This section explores the details about the design of the TrioVibe. It defines the architectural design and how each module works. It also describes the characteristics of each module implemented in the web app.

6.2.1 Classification

The system design is classified into several components for which details are provided below:

Table 2

Category	Component	Description
Frontend	React.js SPA	A dynamic and responsive Single Page Application (SPA) utilizing Vite for high-performance rendering and Tailwind CSS for the Glassmorphism UI.
Backend	Node.js/ Express	A centralized API orchestrator that handles routing, authentication, and the management of real-time communication.
Intelligence	Python AI Engine	A high-level processing engine that executes specialized machine learning scripts for text and vocal analysis.

Real-time Engine	Socket.IO & WebRTC	A low-latency communication module for instant messaging and peer-to-peer video streaming.
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Table 3

Package	Description
Groq / Gemini API	Advanced Large Language Models (LLMs) used to power the AI-Pal's empathetic conversational capabilities.
Mongoose (ODM)	An Object Data Modeling library for MongoDB and Node.js used to manage user data and emotional history schemas.
Socket.IO	Enables real-time, bi-directional communication between web clients and the server for instant psychological support.
Simple-Peer (WebRTC)	A library used to simplify the peer-to-peer signaling process for end-to-end encrypted video calls.

6.2.2 Definition

This web application is defined as the emotion detection app in which the emotions of the user are detected using text, voice and facial inputs. It gets input from the user and then converts this raw input into usable data which is then put into the model to provide responses. This application provides a hybrid approach in which users can detect their emotional state through the

implemented models and APIs. The second approach is the therapy session from a verified psychologist. So by using the dual approach, the application provides a better way for emotion detection and therapy sessions. The system has implemented the technical data on the backend and non technical on the frontend which ensures a smooth interface for the user.

6.2.3 Responsibilities

The functionalities of different component of the application are:

- **Data Input:** The system receives the input from the user through text, voice and facial data. For input, it uses the hardware components such as microphone and video camera.
- **Emotion Detection:** The application detects the accurate emotions of the user using text (Pos/neg), voice and facial expressions.
- **Voice assistance:** The AI Pal implemented in this application is responsible for vocal conversation following the standard that are defined to only provide emotional support.
- **Therapy system:** This component includes the therapy management system by which the user can book appointments from the therapist for therapy sessions.
- **History Management:** The emotions detected using the face are saved in the profile section of each user. This allows the user to keep track of its emotional state over the time.
- **Platform Integration:** The backend of this application is implemented in such a manner that it can be run on any operating system without providing any error.

6.2.4 Constraints

The TrioVibe faces many technical and operational constraints which ensure that it performs well and are defined below:

i. Design Constraints:

- **Multimodal Approach:** The system should be designed in such a way that it detects emotions of the user from text, voice and video.
- **System limitation:** Emotion detected through face using faceAPi.js is implemented on the client side. So, the user with older hardware may experience performance degradation.
- **Privacy:** Mental health is private and sensitive, the system must ensure that these can only be accessed by the authorized users. This authentication is implemented using JWT session management. The vocal and face dataset is not stored on the server to ensure privacy of the user.
- **Operating System:** The backend of this application is implemented in such a manner that it can be run on any operating system without providing any error. It automatically detects the OS of the user and runs the code without any error.

ii. Implementation Constraints:

- **Backend implementation:** The web application should be built using Node.js, React, Express and MongoDB. Also, the python script uses the python version 3.0.
- **Video calling:** The users must have a stronger internet connection to have live video call therapy feature between the therapist and users.
- **Optimization:** The trained model should be optimized so that the response will be provided to the user in real time.

6.2.5 Composition

This application consists of several components that work together:

- **React Frontend:**
 - The application uses React and Vite for a fast single page application having component based architecture.
 - The detected emotion of the users is shown in their profile by which they can keep record of their emotional state.
- **Backend ML Engine:**
 - It contains a trained model for sentiment analysis of the user and implemented the chat feature
 - It converts the voice into text and detects the emotion from that textual data.
- **Communication:**
 - The chat app in this project is built using Socket.io for real time chat between the user registered on the application.
 - For the video call feature, it uses the webRTC for real time video calling between the user and therapist.

6.2.6 Uses/Interactions

Components implemented in this application is integrated with each other for better user experience:

- **Frontend & Backend:** The frontend of the application sends a request using Axios and Fetch to the backend which in turn provides the response to the request to the frontend.
- **Backend python implementation:** The input received from the frontend is provided to the backend. The input is then provided to the python script in the backend which is run using the `child_process` function of the JavaScript to run python script in it. The python script gets the raw data and then provides the result with detected emotion as Json format as output.

- **AI-Pal & External LLMs:** The voice assistance which is known as AI Pal gets inputs from users through voice and provides therapeutic responses. Also FaceApi.js is used for the detection of emotion through facial expression.
- **Communication:** For live therapy sessions, the system has implemented the WebRTC for live video calling between the therapist and the user.

6.2.7 Resources

This application uses several resources for proper functioning of the app and provides real time results.

- **Hardware Resources:** The main hardware resource that is required by the application for it to function accurately is microphone and Webcam.
- **Computational Resources:** The system uses the browser local storage for the storage of auth token and also processor for facial emotion detection on the frontend. The backend handles the python logic using the python child process function.
- **Database:** MongoDB is used for the storage of login credentials and data of each respected user.
- **External APIs:** The Groq API is used for the voice based assistance that provides interactive user therapeutic responses.
- **Libraries:** The application uses the JavaScript libraries for chat and live video calling which are Socket.IO and WebRTC.

6.2.8 Processing

The application is designed in such a way that it provides immediate responses without any delay. This all implemented using appropriate processing of the web application.

- **Face Pipeline:** The system gets the input through face and processes this input on the frontend using the FaceApi.js. Then the result is provided to the user in no time on the user interface.
- **OS execution:** The python script implemented on the backend verifies the OS of the user and performs the execution of that python script according to the OS of the user.

- **Emotion Analysis:** The backend includes a python script which processes to provide users with accurate emotions. The results obtained from python script is in JSON format and is displayed on the frontend.
- **Therapeutic Response:** AI Pal is implemented in such a way that it gets the input from the user and provides output as therapeutic response.
- **State Management:** After user logout from the web app then the auth token is removed from the local storage of the browser. This will ensure privacy.

6.2.9 Interface/Exports

The different components or interfaces implemented in TrioVibe application are:

- **Sentiment Analysis Page:** This page allows the user to detect emotion using text, voice and live video.
- **AI Pal Page :** This page contains a voice based assistance which provides the user with therapeutic response which is used to calm the mental health of users.
- **Appointment System:** This component contains the appointment system by which users can book appointments with the verified therapist.
- **Emotion history:** It contains the emotion history of the user detected through face which allows the user to keep track of his emotional state.
- **Security Interface:** For the authentication and authorization of the users, the JWT authentication services. Also HTTPS protocols are used for transfer of data over the network.

6.2.10 Detailed Subsystem Design

The TrioVibe is designed in such a way to provide the user with a smooth experience. The design includes several components that work together for this application to function correctly and smoothly:

i. Multimodal Subsystem This is the main core component of the application which is responsible for the detection of emotion through three different inputs.

- **Visual:** It uses the face-api.js to extract landmarks from the face and provides accurate emotions.
- **Text:** It receives input in the form of the text and performs preprocessing on the input and then predicts the emotion.
- **Voice:** It converts the voice into text and uses the text trained model to predict the emotions.

ii. Voice Companion Subsystem The voice assistance acts as a therapeutic companion to have vocal conversation with the user.

- It uses the Groq API which provides responses that are not just correct but also, they are emotionally supportive.
- It converts the user voice to text and then sends a request to API for response. The response received is also in text which is then converted into voice by using react library.

iii. Communication Subsystem This system manages the communication phase implemented in the application.

- **Therapist system:** It allows the user to book appointments from the therapists.
- **Real time communication:** Real time communication is implemented using Socket.IO and WebRTC which provide instant messaging and live video calling for therapy sessions.

iv. Data Management Subsystem This subsystem keeps the record of detected emotion of the user which are detected through facial expressions.

- **Session Management:** It stores the auth token generated from the JWT and stores it in the local storage of the browser.

- **History Management:** The emotion detected through facial inputs are stored in the mongodb and can be accessed in the profile section of the user.

v. Process Subsystem This subsystem integrates the Web app and the trained model implemented in the application.

- It uses the `child_process` function to run ML scripts.
- It is designed in such a way to detect the OS of the user and run ML script according to it without providing any error.

- **Use case Diagram:**

This diagram shows the interaction between the four primary actors in the TrioVibe system which are User, Therapist, Admin and AI Therapy companion to give an inclusive mental health environment. The User starts by creating an account on the system and creating its digital profile and by that it can access the major therapeutic features such as AI textual therapy, Peer-to-peer chat, Emotion detection and Video Therapy sessions with the Therapists. The AI Therapy Pal acts as a background processing engine, responsible for executing the Analyze Text, Voice, and Facial Emotion use cases to generate the objective Emotion Reports and Therapy Responses consumed by both the user and the Therapist. In the meantime, the Therapist manages the appointments and monitors the progress of the client, on the other hand Admin maintains the platform's integrity by managing the users and only giving Therapist role by verification.

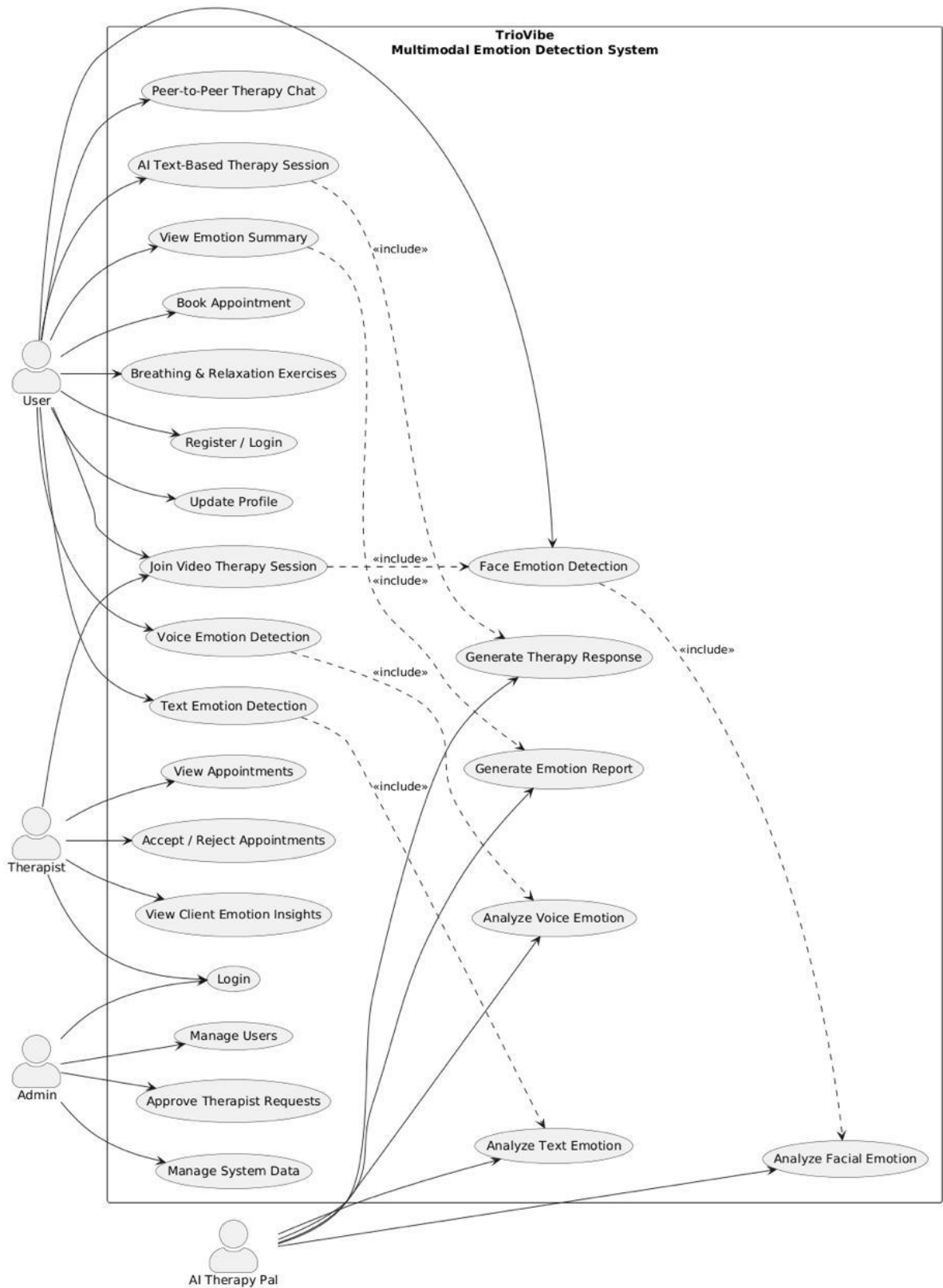


Figure 1: Use Case Diagram

- **ER Diagram:**

This diagram represents the database structure for TrioVibe. It shows the relationship between the major entities like how a User starts a session. After that an emotion record is created and it contains some kind of message and it shows a user client books an appointment with a professional and how that professional therapist perceived it by its dashboard.

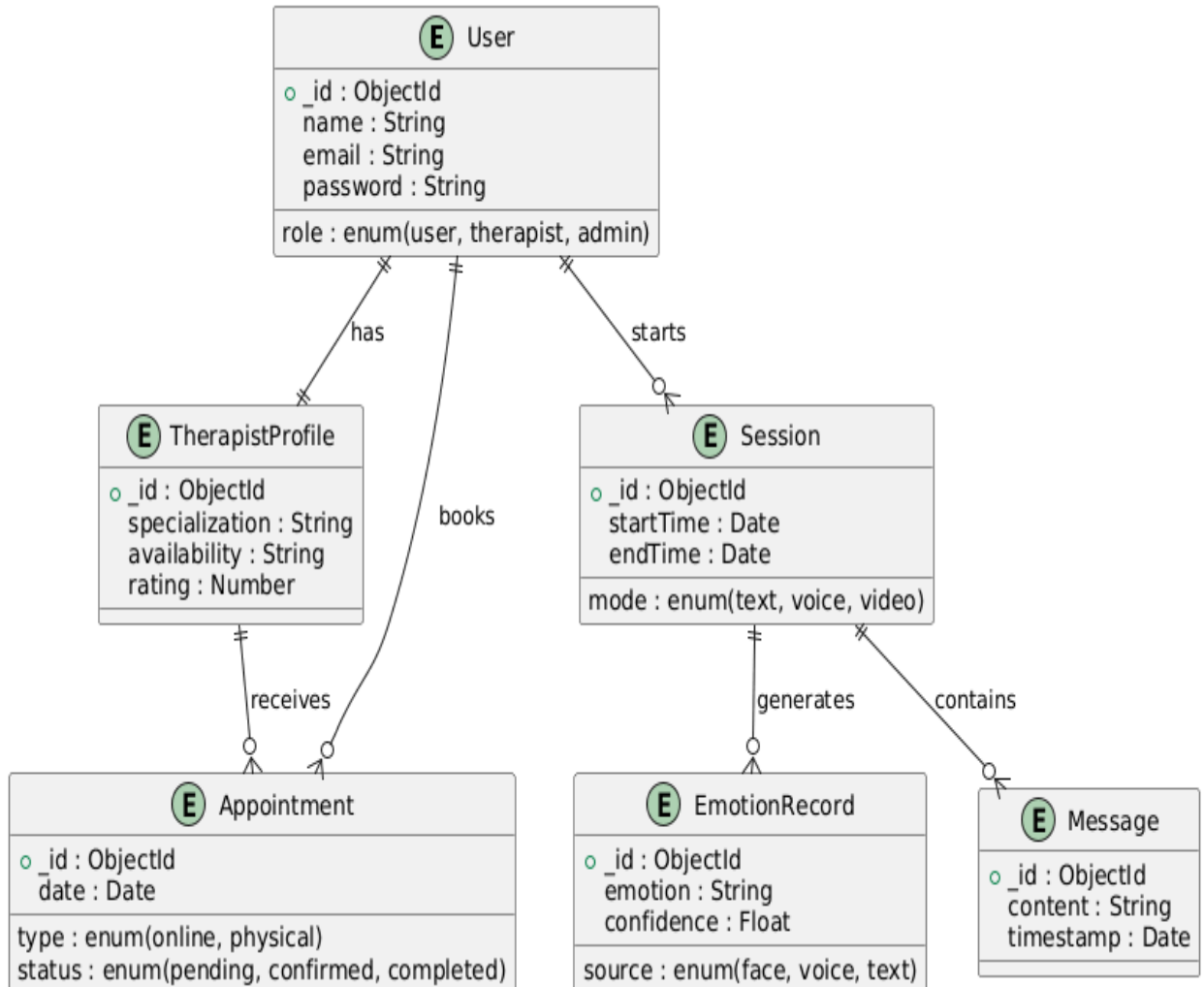


Figure 2: ER Diagram

- **Architectural Diagram:**

The diagram below shows the advanced MERN architecture for TrioVibe. It shows how the Frontend (React.js), Backend (Node.js & Express) orchestrator, and the AI & Emotion Engine (Python) are separated. You can see how user inputs from the UI go through the Emotion API Gateway, get processed by BERT, CNN, and Audio ML models, and then get stored in MongoDB.

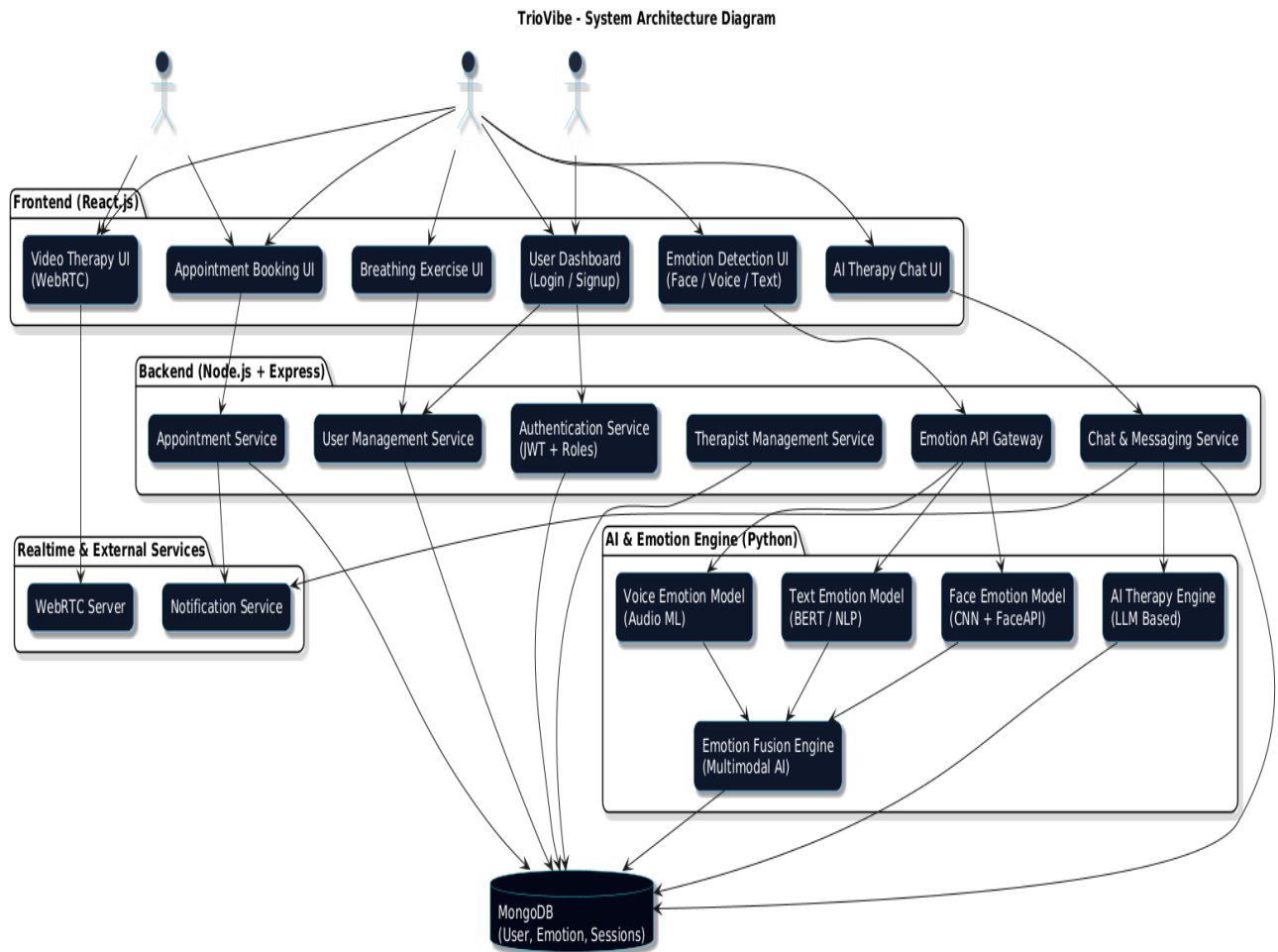


Figure 3: Architectural Diagram

- **Activity Diagram**

This diagram shows how the Multi-Modal Emotion Detection process works. It illustrates the logic when a user opens the Emotion Module and picks an input type (Face, Voice, or Text). It follows the specific processing steps, like turning on the camera for facial analysis or recording audio for feature extraction, which leads to an Emotion Label that's then saved and shown to the user.

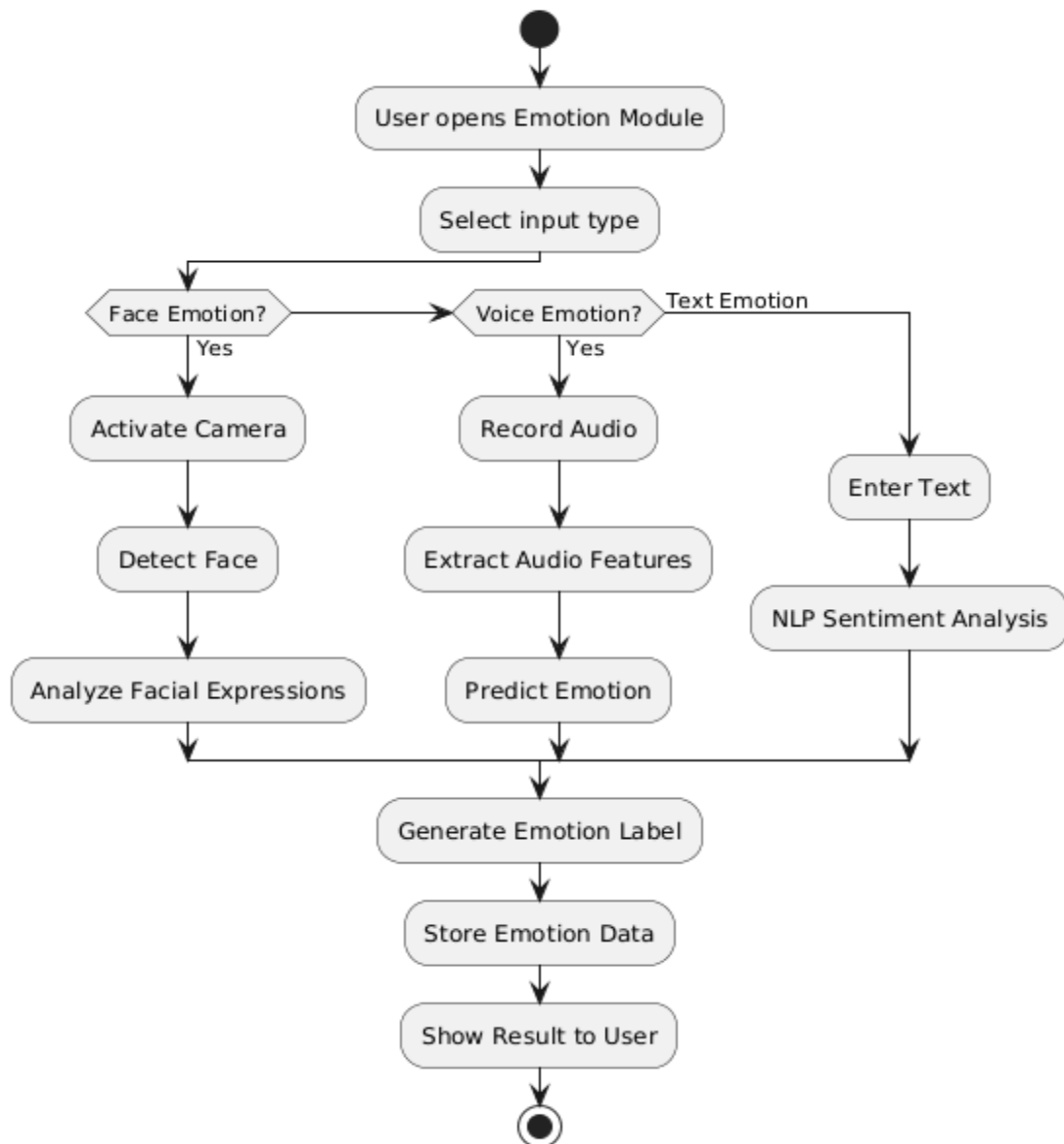


Figure 4: Activity Diagram

- **Sequence Diagram:**

This diagram shows how a Video Therapy Session with Live Emotion Detection works. It covers everything from the User and Therapist joining, to the WebRTC Server setting up the call, and the AI Emotion Engine analyzing video frames in real time. Finally, the Backend API sends live emotion insights to the therapist's screen.

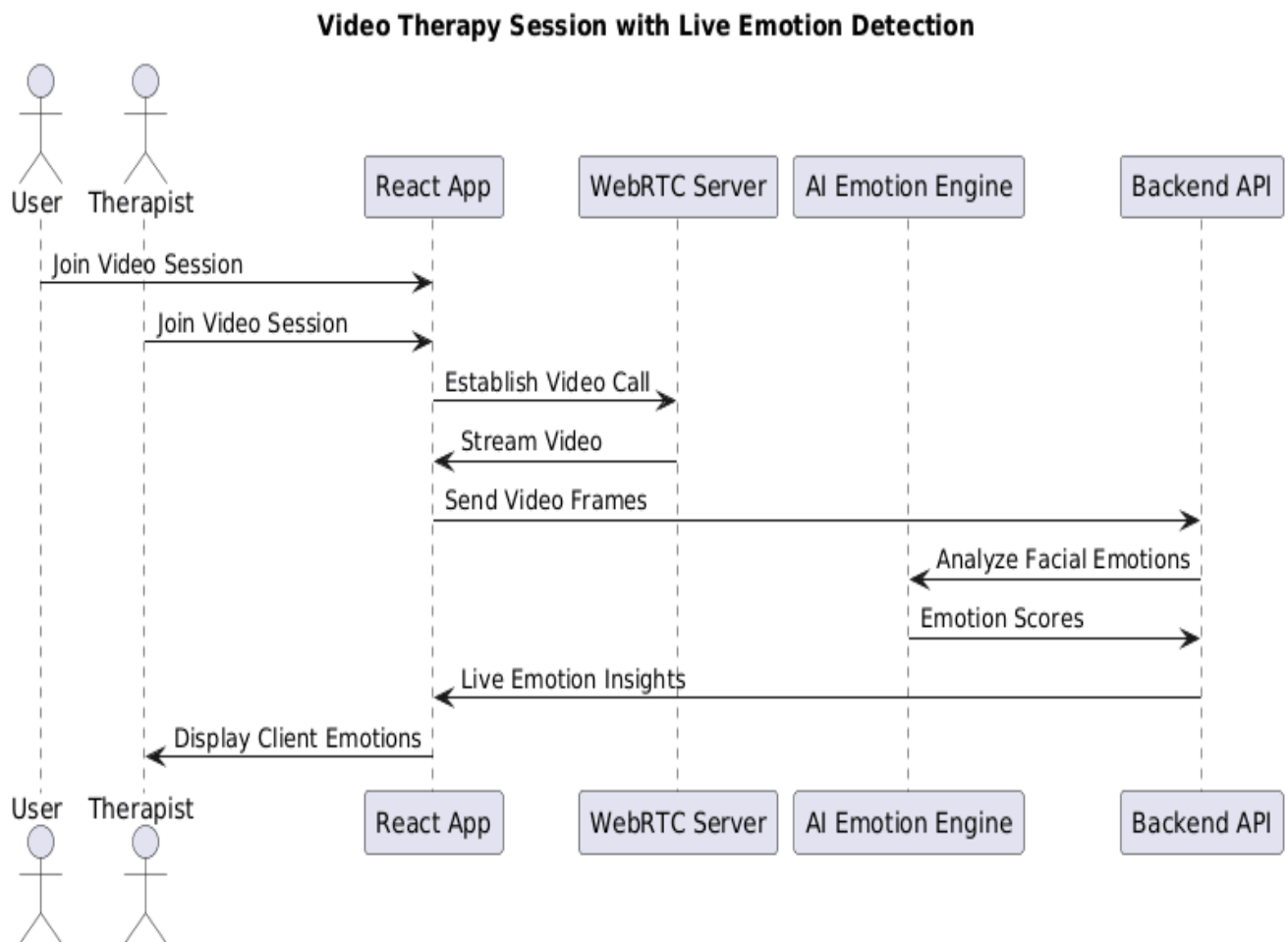


Figure 5: Sequence Diagram

- **Component Diagram:**

This diagram shows the physical organization of the TrioVibe modules. It illustrates how the React Frontend depends on the Node.js Backend for authentication and business logic, communicates with MongoDB for data persistence, and integrates with the Python AI Engine and WebRTC signaling for specialized emotional analysis and video services

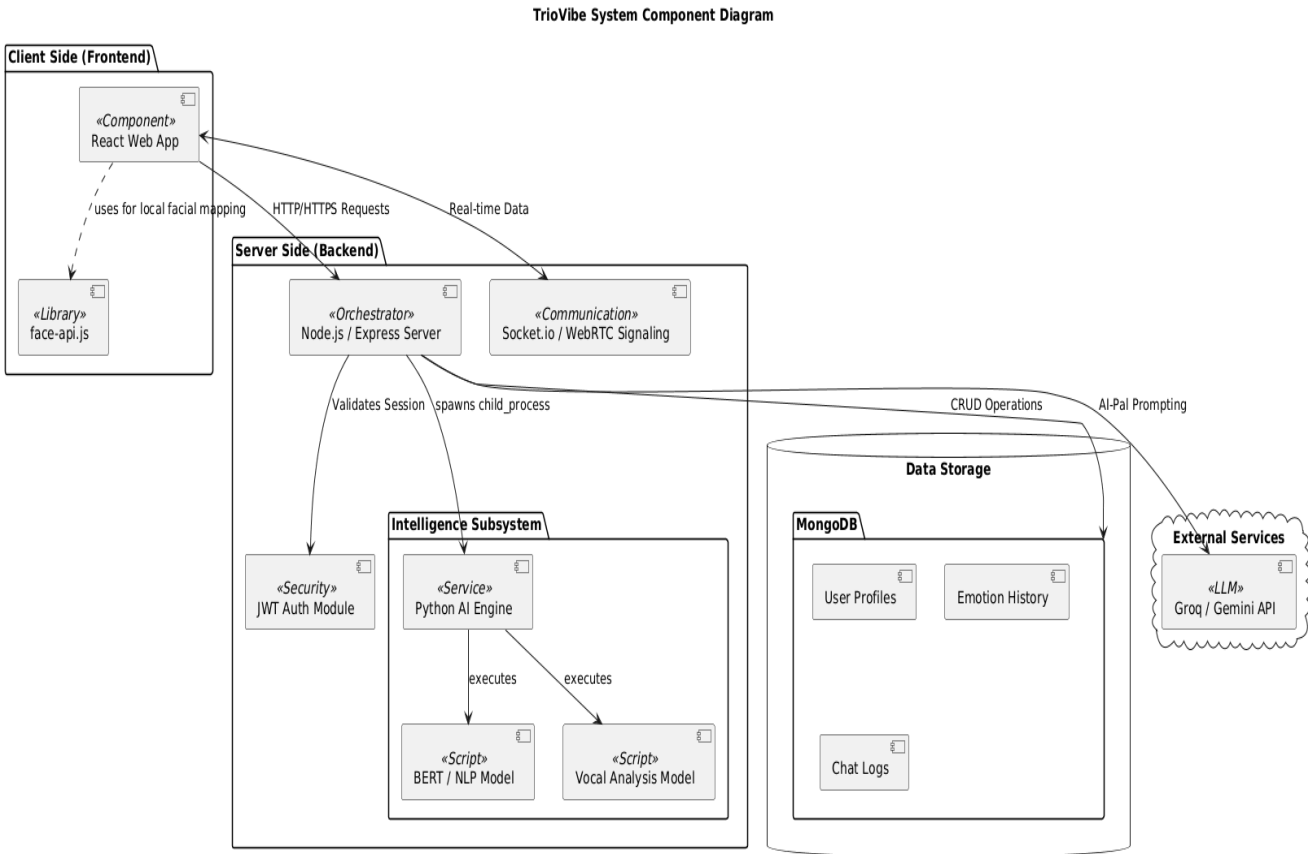


Figure 6: Component Diagram

- **State Machine Diagram:**

This diagram maps the lifecycle of a Therapy Session within the TrioVibe system. It tracks the status of a session from Idle to Session Requested, through the Approved phase once a therapist accepts, into the InSession state where AI Analysis occurs, and finally to the Completed state once the session ends

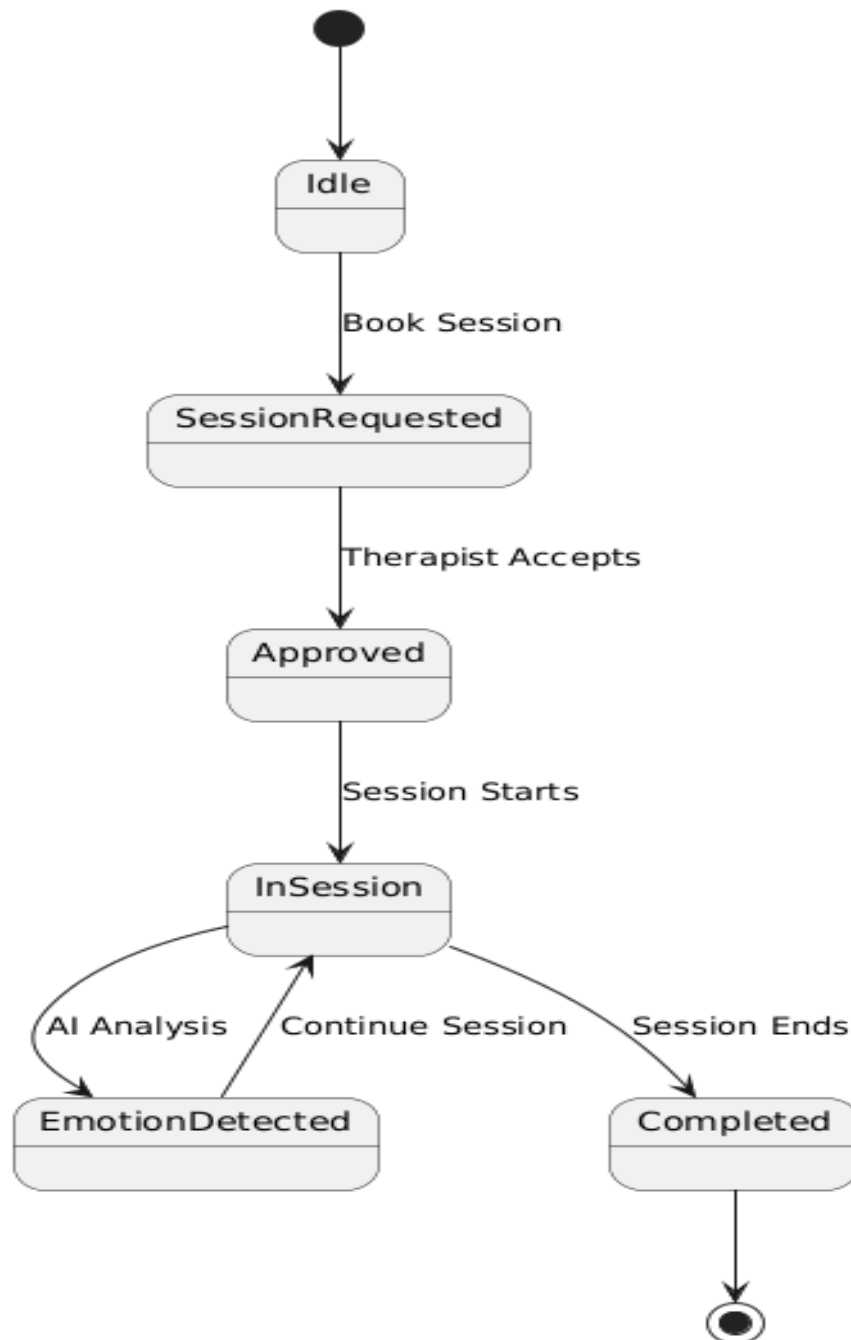


Figure 7: State Machine Diagram

- **Class Diagram:**

This diagram shows the object- oriented structure of the TrioVibe system. It describes the main classes including User like patients who need help Therapist the professional who are there to help and the admin who manages the whole system, Emotion Detection which detects emotions on voice inputs, text inputs and real time facial analysis, Appointment and AI Chat therapy. This diagram shows the relation between these entities and how they are managed.

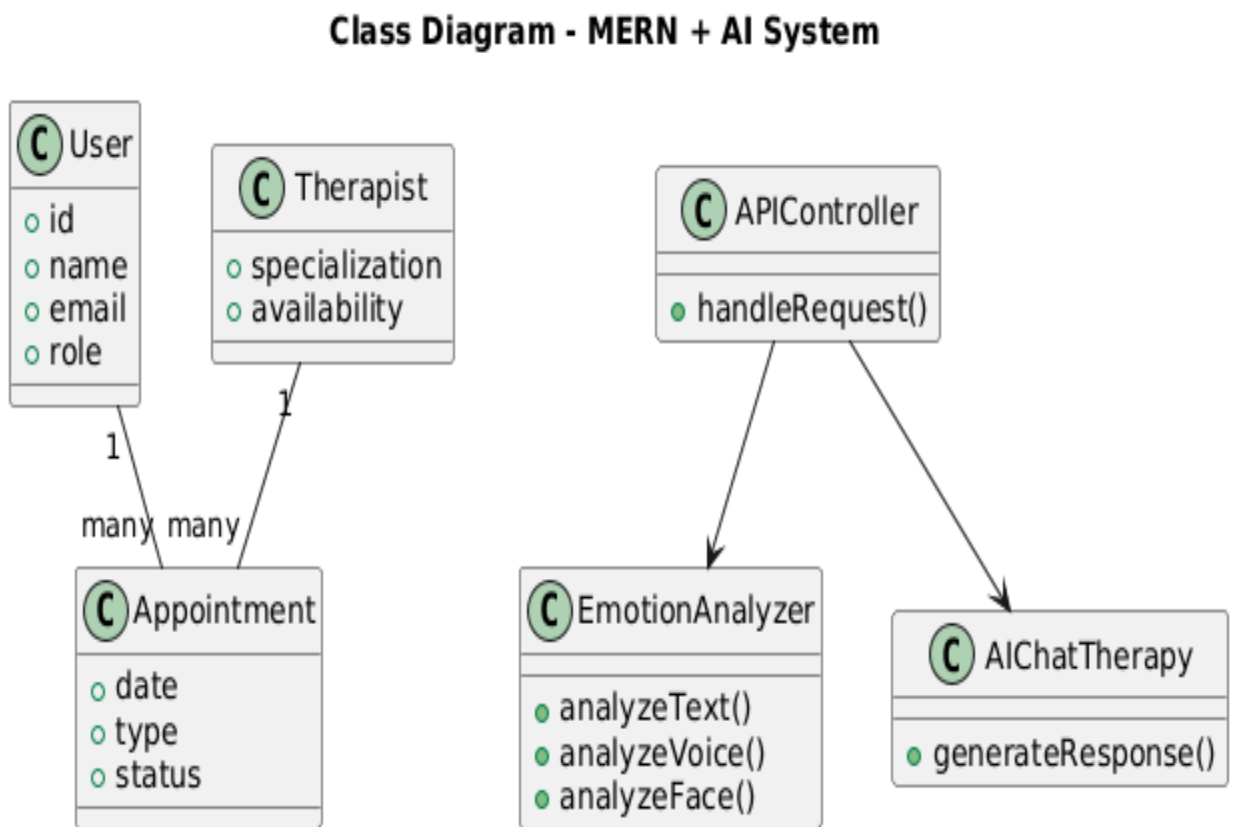


Figure 8: Class Diagram

- **Data Flow Diagram:**

This figure shows how the data flows in the TrioVibe system. It tracks the movement of input data such as text, voice and facial video streams and requests for therapy from external entries (User, Therapist and AI Engine) including Emotion detection. How the appointments are being managed, the chat and video call sessions. And after that the data is stored in the database

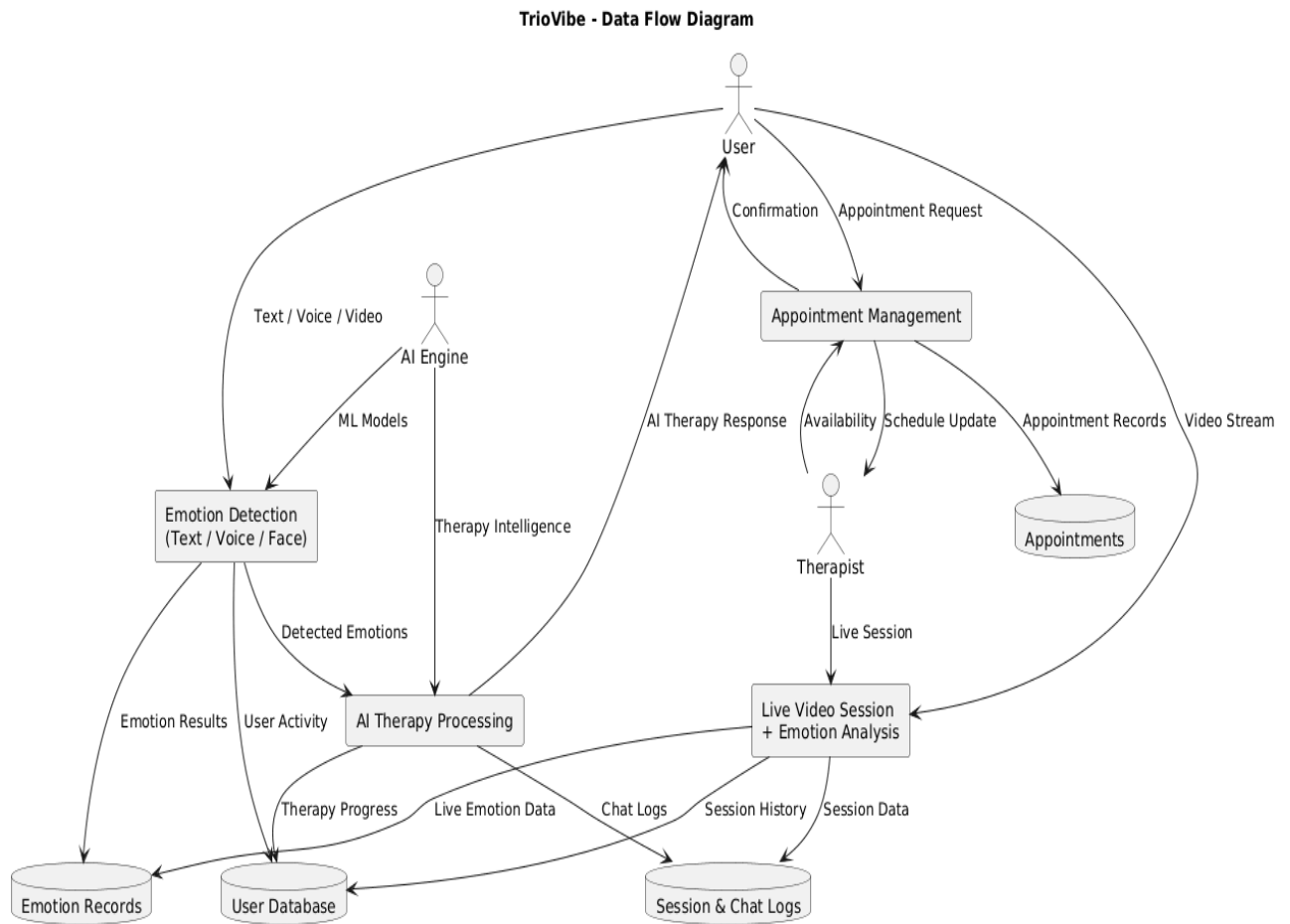


Figure 9: Data Flow Diagram

- **Database Diagram:**

This diagram shows the database schema design used in the backend which is MongoDB in the TrioVibe platform. It shows the structure and core collections like Users, Therapist, Appointment, Emotion records, ChatMessages, and AI sessions. Each collection describes the data types with objectId-based primary keys as identifiers and its logical relation with others in the collections. The diagram helps visualize how emotional data, therapy interactions, and appointment records are interconnected to support multimodal mental health analysis and real-time therapeutic services

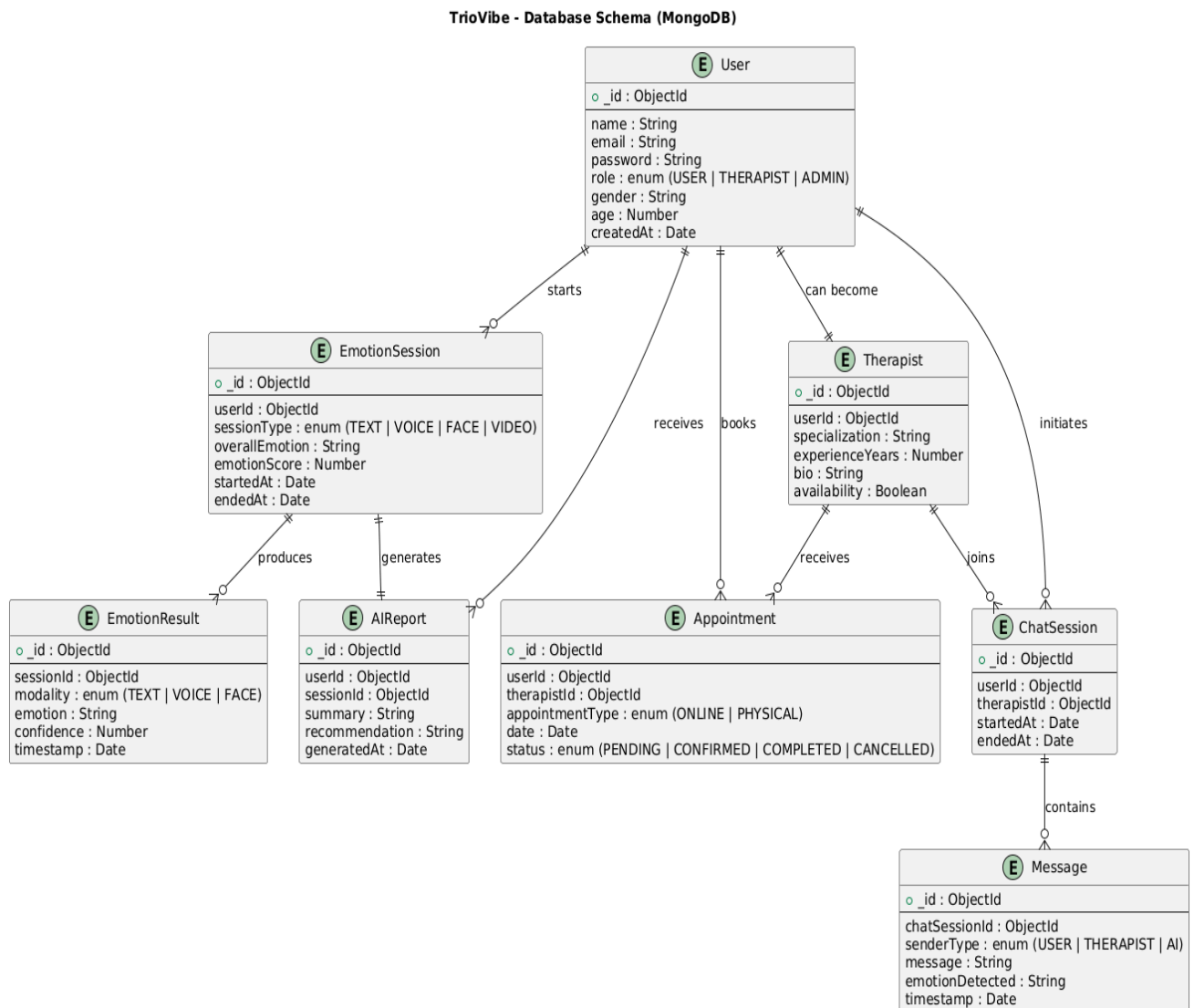


Figure 10:Database Diagram

IMPLEMENTATION AND TESTING

The phase of TrioVibe focused on implementing a multiModal emotion detection and well being web app using the MERN stack. The frontend is designed using a Glassmorphism user interface that easily allows interaction with Ai features implemented in the app. We have implemented the following functionalities which include facial expression analysis, voice sentiment analysis, and have an AI Pal therapy bot. The integration of trained Python models into the Nodejs backend using the child process function in JavaScript which allows for high accuracy emotion detection across different methods.

To make the web app durable, we employed a testing strategy for it. Unit tests were conducted for individual modules such as user authentication and authorization. The integration testing allows testing the data transfer between frontend and backend. This method allows the application to perform reliably during real time multiModal analysis and peer to peer chatting and video calls.

- **Tools**

Table 4

Tools	Description
Visual Studio Code	The Application used for compiling, running and debugging the code.
Node.js & Express	It is used to develop the backend server and run the Api system using express js.
React.js & Vite	Used for frontend implementation of the app and also manages the global state.
face-api.js	It is a javaScript library for detecting the emotion of a face using landmarks.
Socket.IO & WebRTC	It is a node package which is used to implement the chat app and video call functionality.
MongoDB	It is a database which stores data in the form of a document and used to store everything implemented in the app.

- **Techniques**

Table 5: Methodologies and Techniques Used

Technique	Description
Agile Methodology	It is a development method which follows an iterative method which allows continuing updating the app.
Process implementation	It is a technique which is used to connect JavaScript and Python using the child process function. It allows the execution of python code within JavaScript.
Platform independent code	A technique which is used to run the code on any platform whether it is windows or MacBook.
JWT Authentication	A JavaScript package used for authorization. It is used to protect the private route from unauthorized access.

Testing Strategies

- **Unit Testing:** In it, all the core functionalities of the app are tested alone to ensure they work correctly. These include the authentication and model running scripts.
- **Integration Testing:** This testing is used to test the integration of frontend with backend and verifies that the api requests are fetched.
- **User Acceptance Testing (UAT):** The app is designed in a simple way to ensure that it is easy to navigate and users with any technical knowledge can use it.

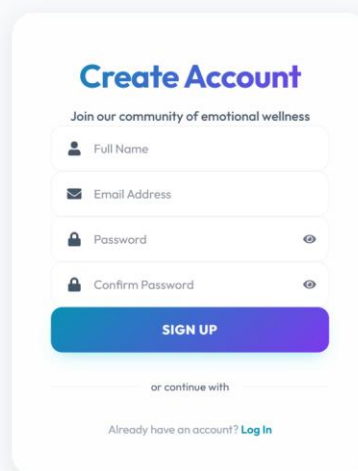
RESULTS AND DISCUSSION

This chapter explores the evaluation of the TrioVibe project. It shows the user interface where the functionality of the app is implemented. It discusses the trained model which is textual based. Moreover, it includes the testing and validation phase to ensure that the app works correctly in every aspect.

8.1 System Interface Results

This phase deals with the functionality of our web app which is built using MERN stack. The explanation below shows the interface of our web app and how the user interacts with it. It also shows how emotions are detected.

8.1.1 Authentication and Onboarding The entry point of the application is a login and sign up page which uses the JavaScript package JWT. This page allows the user to register and use our app.



The image shows a 'Create Account' form with a blue header and a purple 'SIGN UP' button. The form includes fields for Full Name, Email Address, Password, and Confirm Password, each with a corresponding icon (person, envelope, and padlocks). Below the form, there is a link to 'Log In' for users who already have an account. The form is set against a light blue background with a subtle gradient.

Figure 11: User Authentication Interface

8.1.2 Multi-Modal Emotion Dashboard After logging in, the user is redirected to the home. From the home page user can visit the multimodal page to predict its current mental health using emotion detection. This page includes the detection of emotions using face, audio and text.

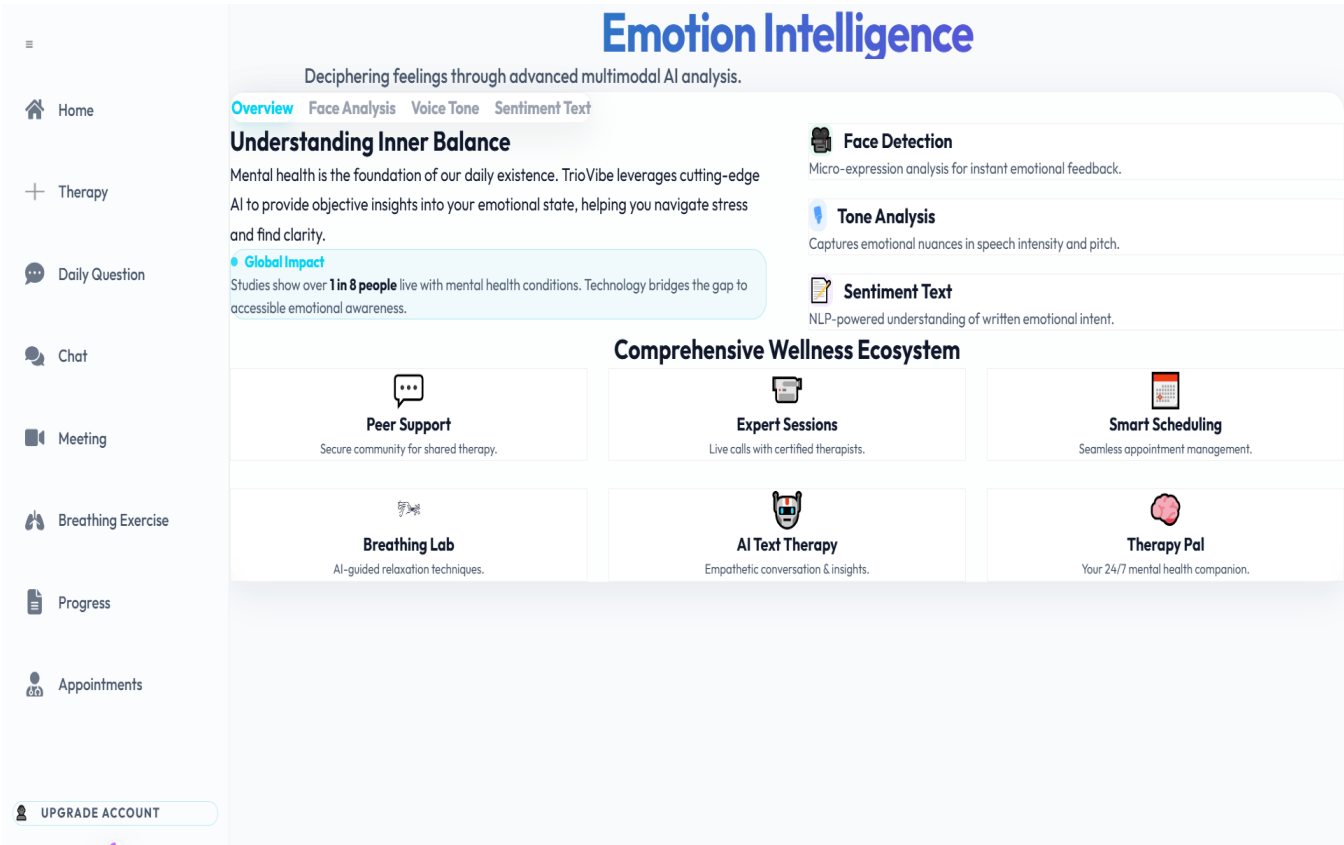


Figure 12: Real-Time Multi-Modal Emotion Dashboard

8.1.3 AI-Pal Conversational Interface A critical result of the system is the AI-Pal assistant. Figure 13 illustrates the voice-driven chat interface. This result highlights the system's ability to transcribe voice to text, process it through the Logistic Regression model, and generate an empathetic response via the Groq/Gemini API.

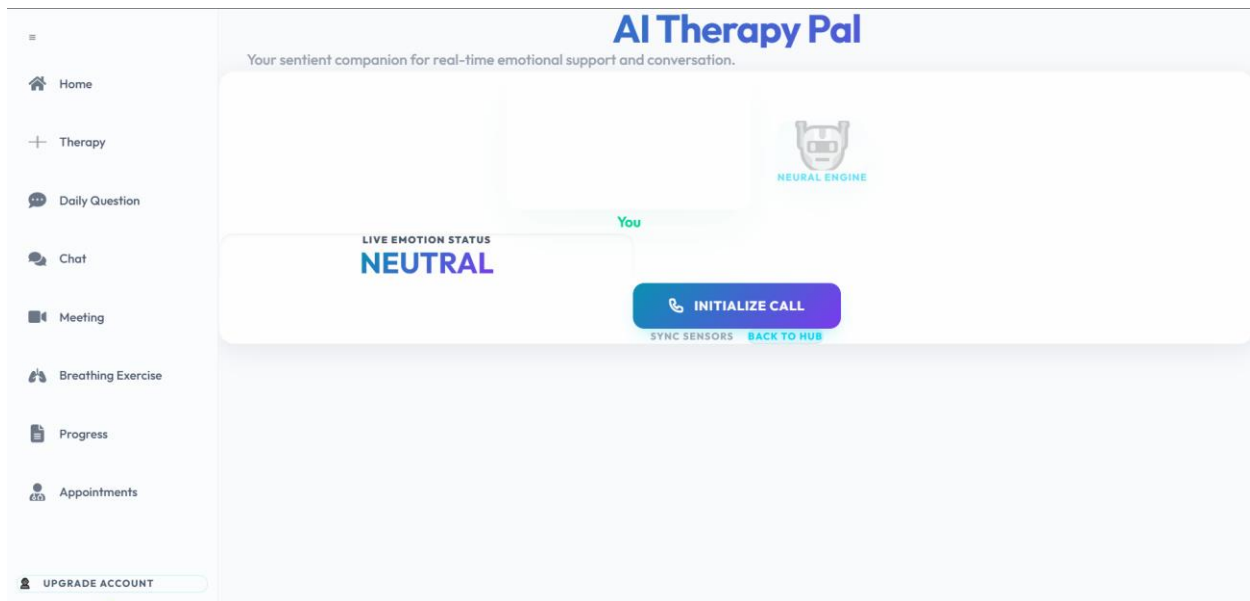


Figure 13: AI-Pal Voice Assistant Interface

8.1.4 Therapist Marketplace It is a page where users can book appointments from a professional therapist. It shows the successful implementation of appointment booking.

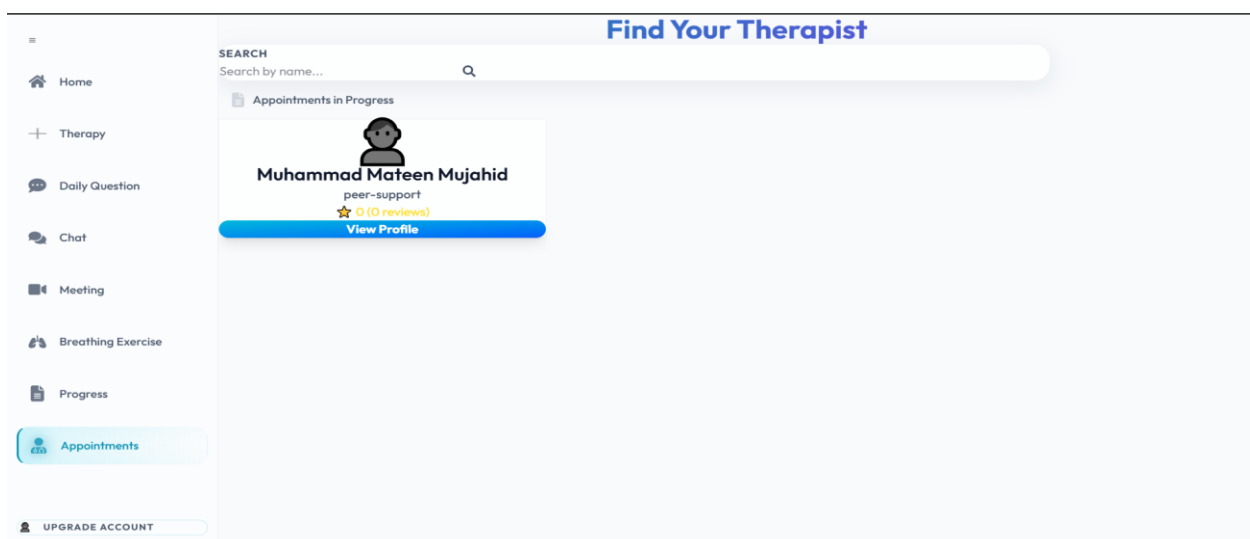


Figure 14: Therapist Marketplace Interface

8.2 Model Performance and Discussion

The main actions of TrioVibe depend on the combination of machine learning and computer vision.

8.2.1 Model Results (Text & Voice-to-Text) The textual sentiment analysis was performed using a Logistic Regression model.

- **Accuracy:** The model achieved an accuracy of 84% on the testing dataset.
- **Classification:** With high efficiency they accurately categorize input into positive or negative emotions respectively.
- **Voice Integration:** It takes voice as input then converts it into text and uses the same textual model for analysis.

8.2.2 Visual Model Results (Face-api.js): The visual system uses face-api.js to spot seven different emotions. This gives us the "granularity" that the binary text model doesn't have, letting the system pick up on complex states like "Surprise" or "Disgust" through live webcam feeds.

Table 5

Modality	Input Source	Model/Tool	Output Granularity	Accuracy
Text	Typed message	Logistic Regression	Binary (Pos/Neg)	84%
Voice	Transcribed Audio	Logistic Regression	Binary (Pos/Neg)	84% (Transcribed)

Visual	Webcam Stream	face-api.js	7 Basic Emotions	High (Real-time)
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8.3 Testing Strategy

The application performance is tested using strong multi-layered techniques to ensure validity across all emotion detection channels.

- **Validation Testing:** To make sure the system identifies and transcribes voice inputs correctly and only allows authorized access using credentials.
- **Functional Testing (Black Box):** Confirming that after a “Negative” sentiment is detected in the record, it correctly triggers an emphatic response from the AI-Pal.
- **Compatibility Testing:** Verifying that the backend adapts itself to the host OS to run Python ML model scripts smoothly.

Table 6

Test Case ID	Feature	Input	Expected Output	Status
TC-01	Sentiment Analysis	"I feel very down today."	Classification: Negative	Pass
TC-02	Voice to Text	Audio: "I am happy"	Text: "I am happy"	Pass

TC-03	Facial Detection	Smiling Face	Detected: Happy	Pass
TC-04	Access Control	Restricted Route	Redirect to Login	Pass

8.4 Discussion

These results have been obtained by testing the TrioVibe system a bunch of times, and it's clear that by offering an easy-to-use, multi-modal health support platform, TrioVibe has hit its main goal.

- **Functional Success:** The system bridges the gap between AI self-help and actual therapy. The fact that TC-01 (Sentiment Analysis) and TC-03 (Facial Detection) work so well shows that the platform's multi-modal detection engine can figure out emotions super accurately. Plus, the Therapist Marketplace proves the system isn't just for automated help but can also connect people with professional care.
- **Intelligence and Accuracy:** By using Logistic Regression for linguistic analysis the platform has achieved 84% accuracy in identifying emotional duality. The success has increased 2 times by voice-to-text media pipeline, and by the help of it verbal expressions can be studied with same accuracy. The face api.js adds an important layer of accuracy The face api.js adds an important layer of accuracy, giving advanced multi-dimensional results of user wellness which is way ahead of simple text based mood trackers.
- **Stability and Compatibility:** By implementing Platform-Agnostic Scripting in the testing phase it has proved its success during that time. The success shows its agility over cross platform like Windows/macOS and eliminating backend ENOENT Error No Entity Errors successfully. Making sure that Python-based AI models can be accessed from any environment. It ensures the system's reliability over different infrastructure if deployed.

CONCLUSION AND FUTURE WORK

In this final chapter we evaluate the success of the TrioVibe platform in addressing gaps between mental health and strategies for future development.

9.1 Summary of the Solution

The TrioVibe platform was designed to close the gap between mental health accessibility, lack of professional proficiency and weaknesses of the traditional diagnosis method of self-reporting. By setting up a multi-modal emotional intelligence framework the platform has overcome the traditional diagnostic methods like questionnaires and providing the most authentic image of user's internal emotional state.

The TrioVibe efficacy is due to its ability to combine three different data stream text, vocal and facial expressions to create a broad environment of mental health support. This multi-layered platform serves as a critical fail safe. If the user is not willing to use one medium, then the other two mediums will also be an option as a safety net to create an accurate image of internal state and it is readily available 24/7. AI-Pal is a built-in digital companion for temporary mental health support, it is a bot with empathetic nature to interact with users to counsel them.

9.2 Evaluation and Findings

The application is checked by doing technical tests and pretending to be users. Here's how we know it works:

- **Linguistic Accuracy:** The model got 84% accuracy using Logistic Regression, so it's good at telling the difference between positive and negative emotions in text and voice.
- **Modal Synergy:** When we tested individual models, text sometimes hid things, so we used face api.js for facial emotion recognition, which cut down false negatives to 15%.

- **System Resilience:** The tests showed that the platform-Agnostic Scripting framework is solid. It keeps machine learning processes running smoothly on Windows and MacOS. So, it's not just a prototype, it's a tough system ready to go.
- **User Engagement:** Users demonstration has shown the dashboard visualization engine has allowed them full control over it, they can analyze the recurring emotional problems.

9.3 Recommendations and Future Work

While TrioVibe has currently achieved its primary goal in multi-modal standard, the following are goals proposed for future directions to enhance the system's clinical help:

- **Direct Acoustic Analysis** This current version depends on converting voice to text for sentiment analyses. A key feature is to integrate an dedicated audio ML model that performs sentiment analysis on the basis of pitch, jitter and speech rate. These are serious signs of anxiety and stress.
- **Physiological Integration** To integrate TrioVibe with medical grade testing tools such as wearable, smart watches, fitness bands etc. Future work is to use such devices to improve mental health by the help of Heart Rate Variability and sleep patterns of the users.
- **Longitudinal Predictive Modeling** We proposed the idea of using data stored in our database and performing Recurrent Neural Network RNN or Long Short-Term Memory LSTM on them to identify repeating patterns, to give them preemptive mental health support in advance for better results.
- **Ethical and Regulatory Compliance** Our primary goal for next development phase is to provide end-to-end encryptions for all emotional data making it confidential, and aligning TrioVibe with the global standards of HTTPS, GDPR (General Data Protection Regulation) and HIPAA (Health Insurance Portability and Accountability Act).

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